Vacuum Cleaner:

```
agent_table = {
    ('Clean', 'A'): 'MoveRight',
    ('Clean', 'B'): 'MoveLeft',
    ('Dirty', 'A'): 'Suck',
    ('Dirty', 'B'): 'Suck',
}


class VacuumCleaner:
    def __init__(self, location='A', room_statuses={'A': 'Dirty', 'B': 'Dirty'}):
        self.location = location
        self.room_statuses = room_statuses # Dictionary to track status of each room


    def percept(self):
        return self.room_statuses[self.location], self.location  # Returns (status of current room,
location)


    def act(self, action):
        if action == 'MoveRight':
            self.location = 'B'
        elif action == 'MoveLeft':
            self.location = 'A'
        elif action == 'Suck':
            self.room_statuses[self.location] = 'Clean'


    def get_cleaning_percentage(self):
        total_rooms = len(self.room_statuses)
        clean_rooms = sum(1 for status in self.room_statuses.values() if status == 'Clean')
```

```
        if total_rooms == 0:
            return 0.0
        return (clean_rooms / total_rooms) * 100

    def get_overall_cleaning_status(self):
        if self.get_cleaning_percentage() == 100:
            return "The entire floor is CLEAN!"
        else:
            return "The floor is not yet clean."

def table_driven_agent(percept):
    return agent_table.get(percept, 'NoOp')

# Simulation
if __name__ == "__main__":
    vacuum = VacuumCleaner(location='A', room_statuses={'A': 'Dirty', 'B': 'Dirty'})
    for step in range(5):
        percept = vacuum.percept()
        action = table_driven_agent(percept)
        print(f"Step {step+1}: Percept {percept}, Action: {action}")
        if action != 'NoOp':
            vacuum.act(action)

        print(f"Current Room Status: {vacuum.room_statuses}")
        print(f"Floor Cleaned: {vacuum.get_cleaning_percentage():.2f}%")
        print(f"Overall Status: {vacuum.get_overall_cleaning_status()}\n")
```

output:

AI.ipynb

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```python
for step in range(5):
    percept = vacuum.percept()
    action = table_driven_agent(percept)
    print(f"Step {step+1}: Percept {percept}, Action: {action}")
    if action != 'NoOp':
        vacuum.act(action)

    print(f"Current Room Status: {vacuum.room_statuses}")
    print(f"Floor Cleaned: {vacuum.get_cleaning_percentage():.2f}%")
    print(f"Overall Status: {vacuum.get_overall_cleaning_status()}\n")
```

```
Step 1: Percept ('Dirty', 'A'), Action: Suck
Current Room Status: {'A': 'Clean', 'B': 'Dirty'}
Floor Cleaned: 50.00%
Overall Status: The floor is not yet clean.

Step 2: Percept ('Clean', 'A'), Action: MoveRight
Current Room Status: {'A': 'Clean', 'B': 'Dirty'}
Floor Cleaned: 50.00%
Overall Status: The floor is not yet clean.

Step 3: Percept ('Dirty', 'B'), Action: Suck
Current Room Status: {'A': 'Clean', 'B': 'Clean'}
Floor Cleaned: 100.00%
Overall Status: The entire floor is CLEAN!

Step 4: Percept ('Clean', 'B'), Action: MoveLeft
Current Room Status: {'A': 'Clean', 'B': 'Clean'}
Floor Cleaned: 100.00%
Overall Status: The entire floor is CLEAN!

Step 5: Percept ('Clean', 'A'), Action: MoveRight
Current Room Status: {'A': 'Clean', 'B': 'Clean'}
Floor Cleaned: 100.00%
Overall Status: The entire floor is CLEAN!
```

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